Using Unmanned Aerial Vehicle Imagery for Estimating River Flow Velocity



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June 30th, 2026

WAGENINGEN
UNIVERSITY & RESEARCH

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Management summary

Accurate assessments of river flow velocity are critical for military river-crossing operations. Currently, applied methods require people to physically go to the riverbank, which is undesirable in conflict areas. This study shows that the combination of using UAV video and a Particle Image Velocimetry (PIV) pipeline is a feasible method to remotely measure flow velocity. An open-source Python implementation was made using the OpenPIV library. This was tested on RGB (normal colour) video footage and validated using ground truth data at two Dutch rivers: the Geul (near Gulpen, Limburg) and the Waal (near Nijmegen, Gelderland). Results indicate that the output from the model systematically underestimates flow velocities and has a Mean Absolute Error (MAE) of 0.35 at the Geul and 0.58 at the Waal and consistent underestimation of flow velocities compared to our validation data at both locations.

To achieve similar results in future cases, we recommend the use of a camera platform with fully known specifications or extractable recording metadata. Secondly, we recommend integrating a method to determine the distance from the UAV to the water surface. During data acquisition, it is important that the water surface is directly or indirectly illuminated. Our method works in direct sunlight and in full shadow but struggles in the transition zone. We identified several opportunities for future research. For one, several processing parameters were not included in our testing. Optimising different parameter configurations could lead to improved results. Secondly, the rivers mentioned in this study do differ in scale and velocity, but testing on other rivers might give better insight into the overall applicability of our method. Third, we discovered a systematic bias in our results. Correcting this would require a calibration method or correction factor to be applied.

At the Geul location, flow velocity with LSPIV from UAV video imagery shows great promise, but suffers from a significant, predictable negative bias. The LSPIV process systematically underestimates river velocity by an average of 0.27 m/s, and this underestimation gets progressively worse in faster-moving water. Because the Shapiro-Wilk test showed the errors are normally distributed and the Pearson correlation showed a clear linear trend, the method should have the potential to be further improved by systematically adjusting the output in post-processing. Measurements and analysis were performed on the Waal as well. We illustrate that our method also shows potential on a wider river. However, the method of validation lacks the reliability to be used for quantitative interpretation.

In short, we deem the use of UAV videos a feasible **o**ption for measuring river flow velocity without Ground Control Points. We hope this work contributes to safer field practices and, eventually, to the preservation of human life.

1. Introduction

Military terrain analysis is an instrumental component in the planning and execution of missions for defence organisations. It helps determine movement, advantages, and possible disadvantages to eventually minimise risks and maximise the chances of a successful mission. Within military terrain analysis, there are five thematic layers: hydro, vegetation, soil, relief, and infrastructure, all contributing to determining the traversability of a terrain. Within the hydrographic layer, one of the aspects involves the assessment of whether a river can be traversed by military vehicles or on foot, either directly or using pontoon bridges. Here, accurate assessment of river flow velocity is critical, as it influences the ability of river crossing by military vehicles or on foot, either directly or using pontoon bridges. Rivers with a width of at least 20 meters are of operational interest, as these typically pose the greatest obstacles for crossing. Currently, such assessments rely on reconnaissance missions in which soldiers physically approach the riverbank and release a floating object to approximate surface flow velocity. This dependency on in-situ measurements puts soldiers at risk in conflict areas and is therefore not preferred. In addition, in-situ measurements are time-consuming, and the ability to gather this information faster than the opponent provides a tactical advantage. Having the ability to remotely sense a river's flow velocity would mitigate this risk and potentially save valuable lives.

In recent years, satellite and UAV (Unmanned Aerial Vehicle) imagery have been used to estimate river flow velocity (Everard et al., 2025; Saletti, 2024). To accomplish this, Large Scale Particle Image Velocimetry (LSPIV) algorithms, combining particle tracking and image analysis techniques, are developed and deployed to automate remote velocity measurement in rivers using either RGB or thermal sensors (Fujita et al., 1997; Eltner et al., 2021; Legleiter et al., 2024). A study from Everard et al. (2025) showed that the flow velocities from eight *SkySat* satellite videos had a deviation of ±5% from Acoustic Doppler Current Profiler (ADCP) and drone-derived values. However, this video imagery of satellites with the required spatial resolution is expensive and has limited coverage. UAVs do provide sufficient spatial resolution, are flexible and have a wide coverage. Yet no transferable open-source workflow for military decision-making specifically has emerged that is operationally validated or widely accepted.

1.2 Problem Statement

Currently, river flow velocity is assessed through in-situ measurements, requiring soldiers to physically approach the riverbank. This poses unnecessary risk to personnel in conflict zones and is time-consuming. While UAV imagery has shown potential for remote velocity estimation, no validated and transferable workflow currently exists for this purpose.

This study therefore investigates whether UAV video imagery, processed through a velocimetry algorithm, can accurately estimate river flow velocity without the need for in-situ measurements. The aim is to assess the accuracy of such a pipeline, determine the minimum spatial resolution required, and identify key constraints related to flight parameters and environmental conditions.

This leads to the following research questions:
(1) To what extent is it possible to accurately determine the flow velocity of surface water using UAV video imagery, and what is the minimum spatial resolution required?
(2) What are the key constraints for the ability to make and use UAV imagery for river flow velocity?

1.3 Research Objectives

The primary objective of this study is to determine whether UAV RGB and/or thermal video imagery, processed through a velocimetry algorithm, can produce surface flow velocities of sufficient accuracy for operational decision-making, as a step towards a standardised approach. Another objective is to identify the minimum spatial resolution required for the detection of surface water velocity, and to quantify the relationship between spatial resolution and detectable flow velocity, as well as optimal UAV flight parameters and conditions for future use. Lastly, this study aims to define key constraints of the methodology, such as the influence of environmental conditions during the UAV data acquisition that may influence the accuracy and applicability of the method.

To answer these questions, UAV flights were conducted over two Dutch rivers, the Geul and the Waal, to acquire RGB video imagery and accompanying field measurements to provide validation data for an accuracy assessment. After this, a Python pipeline combining two LSPIV algorithms was developed to determine flow velocity from the acquired UAV video imagery.

1.4 Deliverables

This project will deliver Python scripts that take drone video footage as input and extract river velocity, developed as a proof of concept to investigate the feasibility of the method, plus a user manual explaining how to use these scripts. Additionally, a UAV and field data acquisition protocol describing the procedures, equipment settings, flight parameters, and conditions required for collecting suitable drone footage in the field will be delivered.

1.5 Reading Guide

The remainder of this report will consist of the following. Chapter 2 provides background on the two LSPIV tools used in this project: RIVeR and OpenPIV. Chapter 3 describes the methodology that was used, this includes: the research areas, the UAV and validation data acquisition procedures at the Geul and Waal, video preprocessing, the GSD calibration method, and the design of the PIV processing pipeline. Chapter 4 presents the results, including the model output, accuracy assessment, and the impact of spatial resolution, flight altitude, and flight duration on velocity estimates. Chapter 5 discusses these findings in relation to existing literature, identifies key limitations, and explores potential alternative approaches. Chapter 6 states the conclusions, and Chapter 7 provides recommendations for future data acquisition, GSD determination, validation, and handling of environmental factors.

2. Background

To estimate river flow velocity from drone footage, this study builds an open-source script based on the functionalities of RIVeR and OpenPIV. Both tools are well-established and validated in many case studies but did not have an open-source python pipeline yet (Ben-Gida et al., 2020; Bandini et al., 2021; Jyoti et al., 2023). This pipeline is designed with the specific conditions of military conflict zones in mind. Several adaptations were made to the core functionality for this context, but it is built on the core principles of RIVeR and OpenPIV. These will be discussed below.

2.2 LSPIV

LSPIV stands for Large Scale Particle Image Velocimetry. PIV (Particle Image Velocimetry) is a technique used to measure velocity in a fluid flow. Large Scale refers to the application for rivers and large water bodies. PIV was originally developed for controlled laboratory settings, where tracer particles are added to a fluid. For rivers, natural surface features such as foam or floating debris act as tracers instead (Fujita et al., 1997). Besides river applications, PIV has also been used in wind tunnel experiments and oceanography (Ben-Gida et al., 2020). For river remote sensing specifically, LSPIV has been applied using both satellite (Everard et al., 2025) and UAV imagery (Eltner et al., 2021; Legleiter et al., 2024; Saletti, 2024).

It works as follows. The frames are extracted from the video, and each frame is subdivided into small regions of interest, called Interrogation Areas (IA) or Interrogation Windows (Raffel et al., 1998). A 2D velocity vector is created for each IA. The corresponding particle pattern is searched for in the next image within the Search Area (SA). A cross-correlation algorithm is applied to identify pattern displacement between IA and SA (Figure 1). This displacement is the average flow velocity in the IA. This cross-correlation is repeated for all IAs in the image, generating a complete displacement field. Ultimately, a velocity field is created, as the obtained displacements are divided by the time interval between two images, giving velocity in metres per second (Massó & Patalano, 2025).

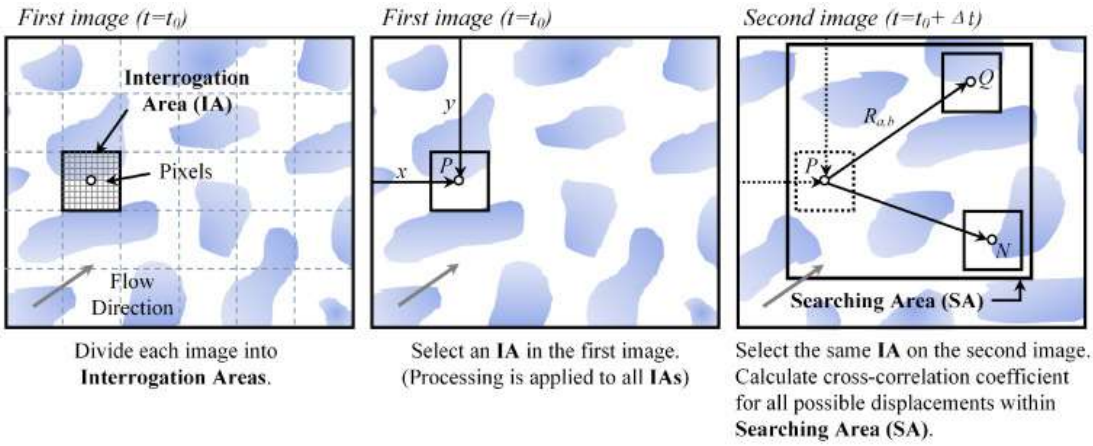


Figure 1. Muste et al. (2008). Conceptualisation of the LSPIV image-processing algorithm, showing the workings of the Interrogation Area and the Search Area.

Cross-correlation is a statistical technique and there are two main methods to apply this. The first is Direct Cross-Correlation (DCC). The approach is computationally expensive, but more accurate than

the Fourier-based method. The second is Direct Fourier Transformation (DFT), which is computed through Fast Fourier Transformation (FFT) (Ben-Gida et al., 2020).

2.3 Rectification of Image Velocity Results (RIVeR)

RIVeR is a toolbox that applies LSPIV to measure river surface velocities from video footage. It was developed in the Centre for Water Research and Technology (CETA) (Ioli et al., 2020 & Kim, 2006). RIVeR was originally built in Matlab but reimplemented into Python. By default, RIVeR uses a larger SA compared to the IA. The objective is to identify river surface features that move faster and more unpredictably than lab particles, meaning larger displacements need to be detected between frames. This process is repeated for the entire image sequence until all images of the video are processed (Massó & Patalano, 2025).

A case study of Bashit et al. (2025) applied LSPIV for image processing in combination with RIVeR for orthorectification of the UAV images. This was applied to the Mungkung River in Indonesia and tested on 3 different flight heights. Even though the results of UAV LSPIV were significantly less accurate compared to terrestrial photogrammetry, UAVs are more flexible in acquiring data, cover a wide area and diminish pixel distortion.

2.4 OpenPIV

OpenPIV is an open-source Python library for Particle Image Velocimetry (Shirinzad et al., 2023). It has been applied in various fields to measure flow velocities from image sequences. In this study, it is used for the core PIV computation between consecutive frame pairs.

OpenPIV offers two variants for the cross-correlation, differing in how the search area is handled: one with and one without zero-padding.

The first, with zero-padding, starts with taking a small patch (IA) from frame 1 and compares it with the same IA on frame 2 to find changes. The search area (SA) in frame 2 is the same size as the IA, the edges are filled with zeros to also look for movement outside the IA. The zeros are made up and can potentially mislead the matching.

The second variant, without zero-padding, uses an SA larger than the IA. Therefore, real image pixel data is used, instead of zero-padded pixels. However, if you choose a window size too large, noise from irrelevant regions is considered as well. By default, OpenPIV uses an equal size between SA and IA, but the function *extended_search_area_piv* offers the option for an SA larger than IA (OpenPIV Guide, 2014). The workflow is visualised in Figure 2.

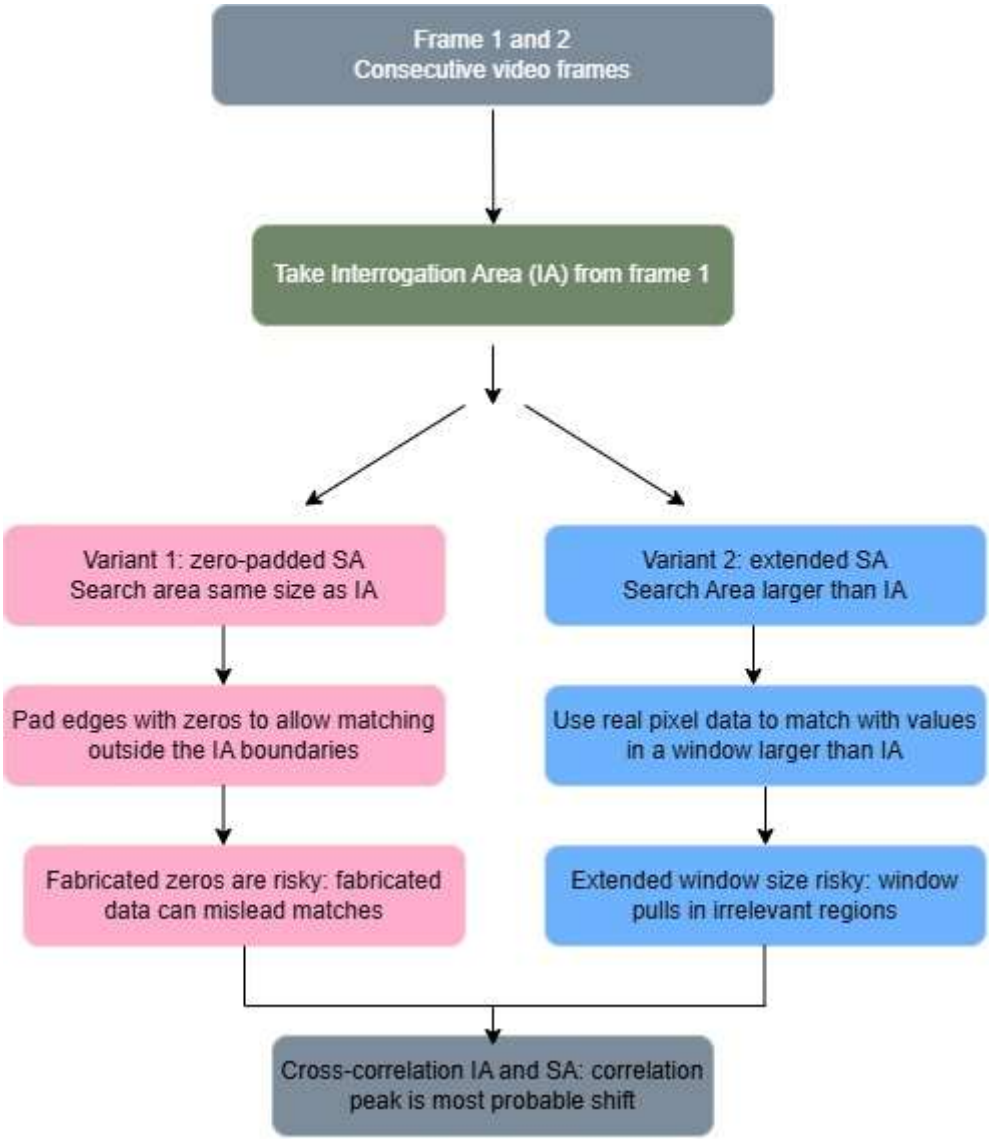


Figure 2. Visualisation of two types of cross-correlation workflows in OpenPIV, showing the path variant 1 takes with zero-padded SA on the left, and the path variant 2 takes with extended Search Areas on the right.

3. Methodology

3.2 Research Area

Study areas were selected based on the following main criteria: sufficient flow velocity to see detectable surface patterns, availability of a suitable validation method (either using a flowmeter of the float-method), accessibility on foot, and permission to operate the UAV at or near the location. Protected nature reservations, other aviation activities and distance to critical infrastructure are limiting factors in performing UAV operations.

For the first research area, three points along the Geul river in southern Limburg were chosen (Figures 3 and 4). The Geul is a relatively fast flowing river that is not deep, making it possible to do flow velocity measurements with a flow meter if the riverbank is accessible. Although the width of this river is smaller than the requested 20 metres, it is a good candidate to assess the feasibility of our methodology.

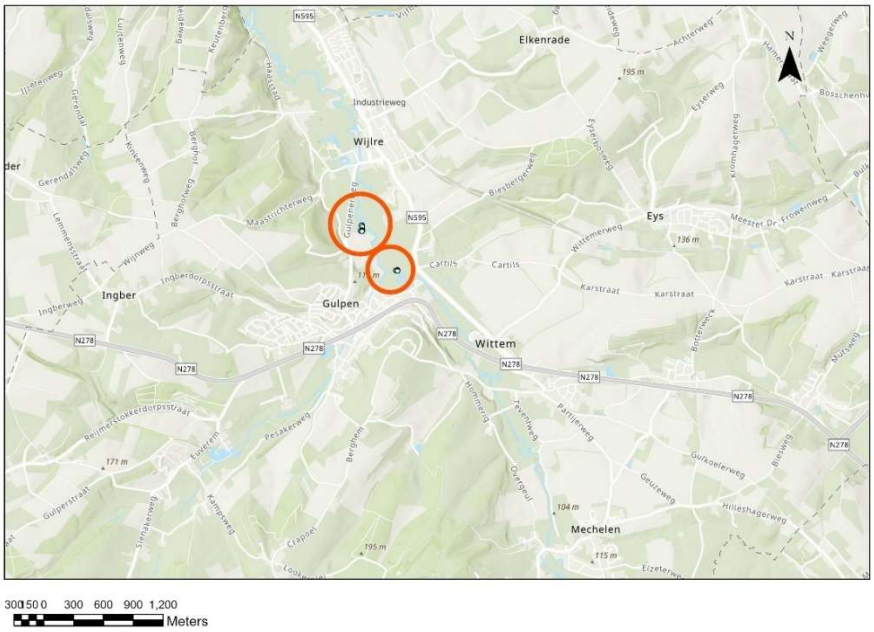


Figure 3. The three research locations at the Geul River in Zuid-Limburg, used for collecting UAV-footage and validation data using a flowmeter.

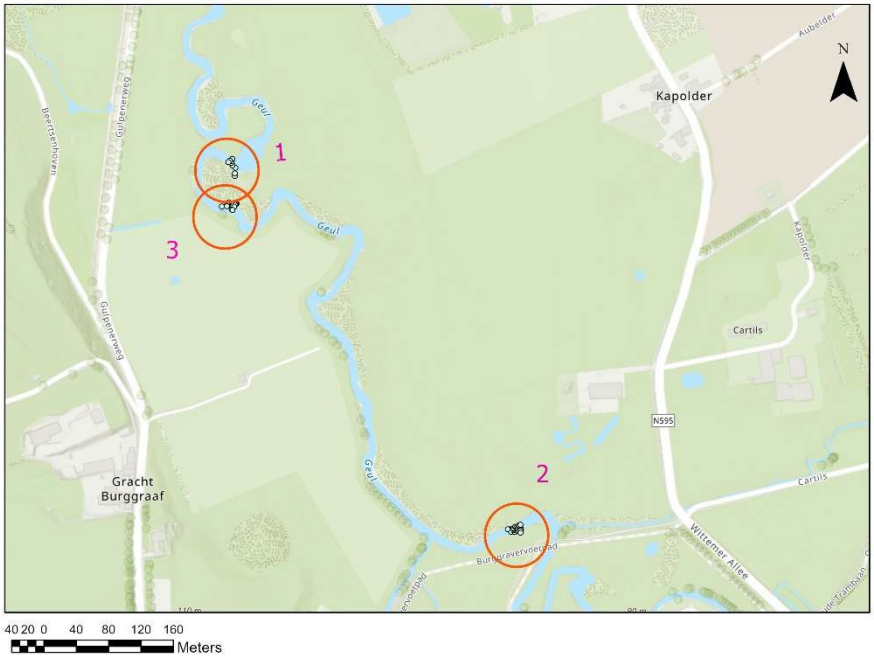


Figure 4. The three research locations of the Geul river in Zuid-Limburg, zoomed in. Used for collecting UAV-footage and validation data using a flowmeter. The points represent the validation points measured with the flowmeter in the river.

The same methodology was applied to a major river with a width larger than 20 metres. The Waal was selected, as it flows relatively fast compared to other large rivers in the Netherlands. For the Waal, validation data was collected using the float method from a bridge, which required the research location to be near one. Based on these conditions, the Waalbrug in the city of Nijmegen was chosen as research location (Figure 5).



Figure 5. The research location at the Waalbrug in Nijmegen. Dots are validation points on the bridge where the float-method was applied. The two most north-eastern dots are the locations where the water height was measured for determining the flight height of the UV. The most north-western point is the location where the UAV ascended and descended.

3.3 Field Data Acquisition

3.3.1 UAV Data Acquisition

UAV data was acquired using the DJI Matrice 4T platform. The UAV acquired natural colour RGB video files at 29.97 fps, as well as thermal data as a compressed RGB video file, at a resolution of 3840x2160 pixels. The goal of each flight was to collect stable, high-resolution RGB and T-RGB video imagery that captures the full river width with spatial resolution and surface texture for LSPIV. The data should preferably be acquired in low wind conditions (<2bf) to minimise the wind's impact on drone stability and disturbance of the water surface.

The drone remained in position at each research location for at least 40 seconds to record the data.

When it was not possible to capture the whole river in a single frame, minimal 10% frame overlap between frames was required. The flight height was location-specific and has been recorded in the metadata. The flight height was location specific because the Geul and the Waal required different flight heights due to the size of the rivers; The Geul is a lot smaller so one can fly lower and still capture the entirety of the river. The differences were also caused by differences in the height between the ascend location and the river, resulting in a slight difference.

For more information on UAV data acquisition can be found in the data acquisition protocol (Annex A) data acquisition plans for the Geul locations and the Waal location (Annex C & D).

3.3.2 Validation Data Acquisition

River flow velocity validation data was collected to compare the results of the LSPIV to in-situ measurements. The methods to do so differ per research location. For the Geul this was done using a flowmeter, and for the Waal locations, the float method was utilised. More detailed information on the validation data acquisition can be found in the data acquisition plans for the Geul locations and the Waal location (Annex C & D).

3.3.3 Environmental Data Acquisition

Several environmental parameters were recorded for each flight to determine possible environmental constraints regarding the acquisition of the UAV imagery for the LSPIV. The following was recorded:

- Wind speed (target < 2 Bft)
- Cloud cover
- Time of acquisition for sun angle/illumination
- Water surface texture
- Presence of shadows (e.g. from vegetation)

More detailed information on the environmental data acquisition and the recording of this data can be found in the data acquisition protocol (Annex A) data acquisition plan for the Geul locations and the Waal location (Annex C & D).

3.4 Data Preprocessing

3.4.1 Calculating Pixel Size using RIVeR Orthorectification

The software of RIVeR was used to calculate the pixel size or Ground Sampling Distance (GSD) of the imagery. The pixel size is a crucial component in the methodology to be able to convert the pixel displacements into velocities when using an LSPIV algorithm (Massó & Patalano, 2025).

For this, two Ground Control Points (GCPs) needed to be visible within the UAV imagery frame. In the field, the point coordinates of these GCPs were recorded using a mobile Global Navigation Satellite Systems (GNSS) device with Real-Time Kinematic (RTK) technique, providing centimetre accuracy. Using this, the real-world distance between these two GCPs could be computed by measuring the distance between them in ArcGIS with centimetre accuracy from the RTK-GNSS coordinates. In RIVeR, the same two points were manually selected, and the program calculates the Euclidean distance between the corresponding pixels.

The pixel size, or GSD was then computed within RIVeR as:

$$GSD = \frac{Real - world\ distance\ (m)}{Pixel\ distance\ (px)}$$

This GSD is equal to the ground distance of a single pixel in the imagery. This factor is used to convert the pixel displacements into velocities as metres per second (m/s).

3.4.2 Video re-encoding

During initial testing, the RIVeR software identified the video metadata incorrectly; a 960x540 pixel resolution and a framerate and frame count that did not correspond to the expected file properties. This mismatch was caused by the DJI storing two video streams; a high-resolution 4K stream (3840x2160 px), and a lower resolution preview stream (960x540 px).
To resolve this issue, the UAV video recordings were re-encoded into a single video stream consisting of the 4K resolution video. To execute this, videos were processed using the HandBrake software. The footage, originally encoded according to the H.265 standard, was re-encoded to H.264 (x264). Within this process, a constant framerate of 29.97 was selected to ensure a constant time interval between frames. The encoding was performed with a quality compression mode controlled through the Rate Factor (RF). The RF controls the trade-off between visual quality and file size (compression), where lower RF values produce higher quality and less compression, while higher RF values produce lower quality and more compression. For the UAV video processing, the default RF value of 26 was selected. This value results in moderate compression, while maintaining sufficient visual quality and texture detail for LSPIV analysis, aligning with the recommended RF value for 4K imagery stated by HandBrake documentation (HandBrake Team, n.d.).

There exist other software packages that provide the same functionality as Handbrake. FFmpeg, for example, was used for video metadata extraction, but can also be used for video re-encoding. The resulting processed videos retained the original spatial resolution and framerate while eliminating the embedded low-resolution stream and reducing the file size. No preprocessing was performed on the videos destined for the LSPIV analysis pipeline, as RIVeR was not used in that process.

3.4.3 Alternative Method to Calculate Pixel Size

Initially, the determination of the pixel size in our methodology was dependent on the distance extracted from the Ground Control Points (GCPs) placed in the field. However, this approach is less ideal for conflict zones, where minimising sending people to exposure areas to assess river velocity and conditions is crucial.

Alternatively, the pixel size or GSD can be determined with a multi-height UAV flight experiment. This can be done remotely before flying the drone over the river. In our case, a multi-height experiment of 10-metre intervals was conducted. The UAV shortly filmed an area with GCPs in 4K resolution and fixed camera settings. A set of GCPs was visible in the footage while doing this. The location coordinates of these points were recorded with the mobile GNSS device with RTK technique to extract the real-world distance between these points. Afterwards, for each altitude, the corresponding pixel distance between the GCPs was extracted in RIVeR as described in section 3.3.2 and shown in Figure 6. The exact distances between GCP points can be found in Appendix I.

Consequently, using the data acquired during the experiment, a linear regression can be applied to determine the relationship between pixel size and flight height (Figure 7). This points at a linear relationship as intrinsic camera parameters remain constant across flights. The relationship between the pixel size (GSD) and flight height is as follows:

$$GSD\ (m/pixel) = 0.00023833 * height\ (m)$$

This formula only has height as input; it assumes a resolution of 3840 x 2160 pixels. To down sample it to another resolution, the outcome must be multiplied by a scaling factor.

The factor is calculated as follows:

$$Scaling\ factor = \frac{image\ height\ at\ 4K\ (2160)}{image\ height\ at\ target\ resolution}$$

For example, a 720p video has a pixel height of 720. The pixel height of 4K video is 2160. So, the scaling factor becomes 3. Besides that, the formula can be applied to all footage made using this specific camera platform. Using a different sensor will produce different results. The experiment can be repeated for other drone and sensor combinations to determine the relationship between pixel size and flight height. The sensor-specific formula can be used to extract the pixel size if the flight height (and scaling factor) is given.

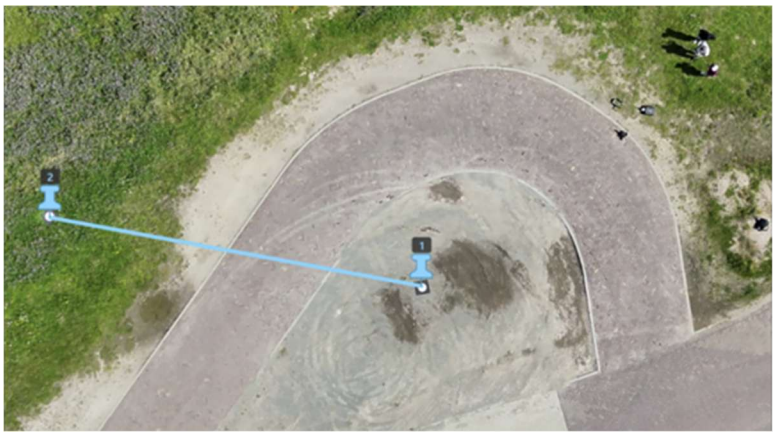


Figure 6. Distance determination between two GCPs in RIVeR's GUI. For the procedure of determining the distance, two GCPs are selected in the frame captured by the UAV, and the software computes the pixel size based on the provided distance between them along the connecting line (blue).

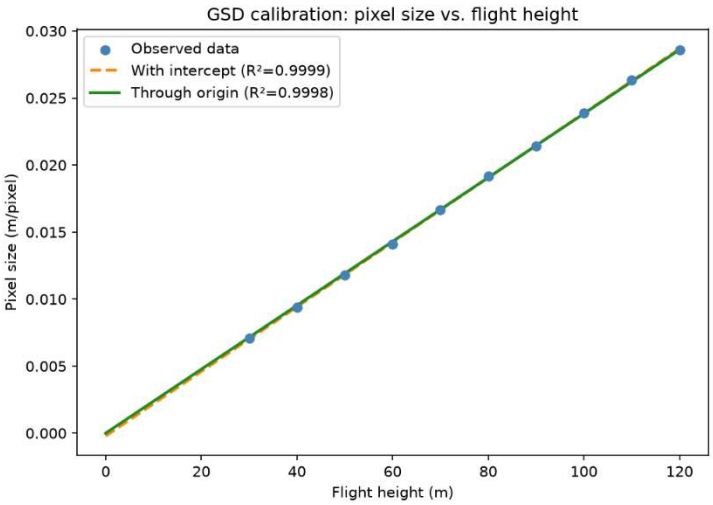


Figure 7. Linear relationship determined using the UAV and ground control points. The distance between two ground control points was determined, and the drone made footage of the points at several heights. Resulting in a linear relationship which can be used to determine the GSD based on flight height.

3.5 PIV Processing Pipeline

The PIV pipeline automates the extraction and validation of the surface flow velocities from UAV-based video recordings. The pipeline, written in Python, integrates preprocessing, PIV computation, georeferencing, and statistical evaluation. This section will describe the pipeline. A summary of the Python pipeline is outlined in Figure 8.

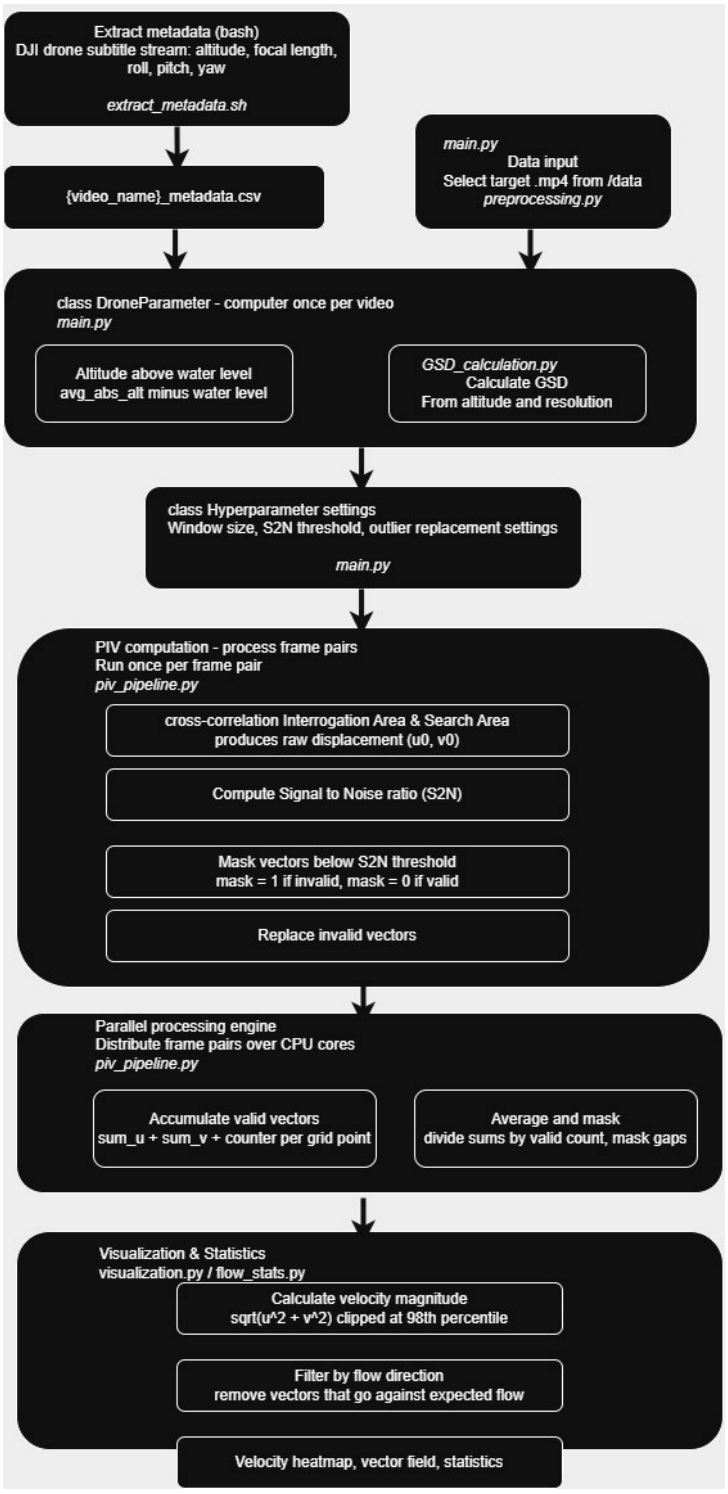


Figure 8. Primary processing pipeline. The image shows the most important steps in drone data processing, validation data processing and accuracy assessment through analysis of differences and statistical tests.

3.5.1 Data Input

The first step of the PIV pipeline consists of selecting a .mp4 file from the /data directory for PIV processing. This step searches for and selects a single .mp4 file as the pipeline is designed to process one video at a time.

If no video is found, or if multiple videos are detected in the directory, this step raises an error to ensure that the pipeline receives only one .mp4 file to process. When a single .mp4 file is found, its path is returned, so that the pipeline knows which video to process.

3.5.2 PIV Computation

In piv_pipeline.py, the LSPIV is performed. This section describes the functions within the pipeline.

The function *process_frame_pair()* computes the PIV analysis on one pair of frames.

It computes the raw horizontal and vertical displacement values (*u0, v0*) based on the comparison between the interrogation windows within the frame grid and search area. Each displacement value is a correlation score reflecting how well a pattern of pixel intensities is detected in the analysis window from frame A matches with the shifted location in the search region in frame B.

Subsequently, the signal-to-noise ratio (SNR) is computed with the peak2peak method, finding the highest correlation value and the second-highest value anywhere in the correlation map. The SNR is de highest correlation value divided by the second-highest correlation value. Based on the signal-to-noise threshold value, vectors with a SNR lower than the threshold are considered invalid and masked out. This is done by marking these vectors with a value of 1. Vectors that are higher or equal to the threshold are considered valid and are unchanged (mask = 0).

Invalid vectors are then replaced using local spatial averaging. For each invalid vector, the algorithm computes the mean of valid neighbours within a kernel region defined in the configuration parameters. This process is iterated (up to *REPLACE_MAXITER* times) because some invalid vectors initially have no valid neighbours; after the first pass, newly filled vectors allow other vectors to be filled in subsequent passes. This results in a grid of vectors.

After outlier replacement, the displacement vectors are scaled to physical velocity units using the known scaling factor and time separation DT. Finally, the coordinate system is transformed so that the origin is at the bottom-left and the orientation matches standard scientific conventions.

The function returns the velocity component in the x-direction, the velocity component in the y-direction (v) and the mask for that frame pair.

3.5.3 Parallel Processing Engine

In the parallel processing engine, all frame pairs in the video can be processed by distributing them across all available CPU cores. First, all frames in the videos are converted into grayscale and stored in a list. For these frames a coordinate grid is computed for later interrogation window division.

Furthermore, frame pairs are created for which computation will be done. This processing engine is designed in a way that each worker executes the function in step 2 (*process_frame_pair*) on a different frame pair. This allows many frame pairs to be processed simultaneously, reducing the overall computation time. As each pair is processed, u and v vectors are accumulated into a grid, and the valid ones are added to sum vectors (*sum_u, sum_v*). To count how many valid vectors are added by each grid point, a counter is included.

After all frame pairs are processed, the final velocity field is calculated by dividing the accumulated sums by the number of valid contributions at each grid point. Grid points with insufficient valid data are masked out.

3.5.4 Configuration Parameters

The pipeline contains eleven configuration parameters. These parameters, their description, and function in the pipeline are summarised in Appendix II.

3.5.5 Visualisation

The velocity magnitude for each grid cell was calculated as $\sqrt{u^2 + v^2}$, using the horizontal (u) and vertical (v) velocity components returned by the PIV pipeline. Only valid vectors (mask = 0) were retained for visualisation. To prevent extreme values from dominating the visualisation, the highest velocity values were clipped with a percentage threshold of 98%.

Using local (x,y) coordinates representing the pixel count, the velocity magnitudes were plotted on a 2D velocity grid. In addition, vectors that do not match the general flow direction of the river are filtered. The general flow direction (e.g. “right” or “down”) translates to a criterion for the u or v component of the vector. For example, when the river flows “right”, all vectors with a u-component equal to or lower than 0 are filtered out.

The velocity heatmap was plotted with the *matplotlib.imshow()* function. A colour bar was added to indicate the velocity from 0 to the clipping threshold. The vector field plot was added with the *display_vector_field()* function from OpenPIV.

Within the processing loop, summary statistics were calculated: minimum, mean, median, 95th percentile and maximum velocity.

Vector values marked invalid (mask = 1) were excluded from visualisations.

3.6 Data Analysis & Assessment

3.6.1 Comparison with Field Measurements

The accuracy of the LSPIV-derived surface velocities was evaluated using the field measurements collected at each research location.

To do so, a csv file containing the name of the point location (integer), geographic coordinates (lat, long), and the flow velocity (flow_speed) was loaded in. To enable spatial comparison between the field measurements and the velocities derived by LSPIV, the velocity raster was georeferenced using the mean latitude and longitude of the UAV, altitude above the water surface, gimbal yaw, and camera specification.
Each validation points was then mapped into the georeferenced velocity raster map. Points within the extent of the raster map extracted the corresponding PIV surface flow velocity. For each point, the difference between the PIV velocity and the field measured velocity was computed.

Moreover, all validation points from the research locations were merged into one dataset for an overall accuracy assessment. Descriptive statistics were calculated, including bias (mean error), median error, standard deviation, and mean absolute error. To test for systematic bias, a one-sample t-test and a Wilcoxon signed-rank test were conducted. Lastly, the proportional bias was evaluated using Pearson correlation between the observed velocities and residuals.

3.6.2 Flight Parameter Analysis

To explore the pipeline's ability to accurately estimate surface flow velocities, the effects of different spatial resolutions, and flight durations were assessed.
The objective of testing the methodology for these different flight parameters is to determine how they influence the accuracy of the flow velocity estimates and eventually determine minimum video and parameter requirements.

To evaluate how spatial resolution influences the performance of estimating river flow velocity, the original 4K UAV imagery was systematically downsampled to a set of lower resolutions. Downsampling reduces both the spatial resolution and pixel count of the imagery, mimicking a situation where imagery would be acquired from higher altitudes or when using a sensor with lower resolution.
Downsampling was performed using bilinear interpolation. With this method, the output pixel is computed as a weighted average of the four nearest pixels in the input image i.e. a 2 x 2 neighbourhood. The different resolutions were created by using a scaling factor. The scaling factor specified how far apart the sampling points lie in the original image. This means that larger scaling factors correspond to sampling regions positioned further apart, meaning each output pixel represents a larger region of the original image, contains more averaging, and therefore has a lower image resolution.
The following spatial resolutions were tested:

- 2160 pixels: 3840 x 2160
- 1080 pixels: 1920 x 1080
- 720 pixels: 1280 x 720
- 480 pixels: 854 x 480
- 360 pixels: 640 x 360

All different spatial resolutions were tested with a video length of 40 seconds with the same workflow as described in Section 3.4.

To assess how the length of the UAV recording influences the velocity estimates by the PIV algorithm, the original videos were clipped to a set of shorter durations. Flight duration directly affects the number of frames available for the velocimetry algorithm to analyse, and through this the availability of tracers to estimate flow velocity.
Clipping was done by extracting the first N seconds of each video, ensuring that the videos of all durations were derived from similar conditions (e.g. lighting, slight changes in UAV positions). This was only done for the videos in 4K (2160 pixels). Eventually, the clipped videos were processed using the same PIV workflow as described in Section 3.4.
The following flight durations were evaluated:

- 5 seconds
- 10 seconds
- 15 seconds
- 20 seconds
- 30 seconds
- 40 seconds

4. Results

4.2 Model output

The model produces the following final output to assess the flow velocity of the Geul river: a velocity grid (heatmap), vector map and RGB image (Figure 9, 10 & 11). This is the primary output of the model when used in a 'real' application. The velocity grids are based on the 4k resolution, 40 second videos for maximum detail. The vector field are based on 720p resolution, 40 second videos as these provide the 'clearest' visualisation of the vector field, covering the image with frequent datapoints while minimising cluttering. The vector field is natively produced at the same resolution as the velocity grid.

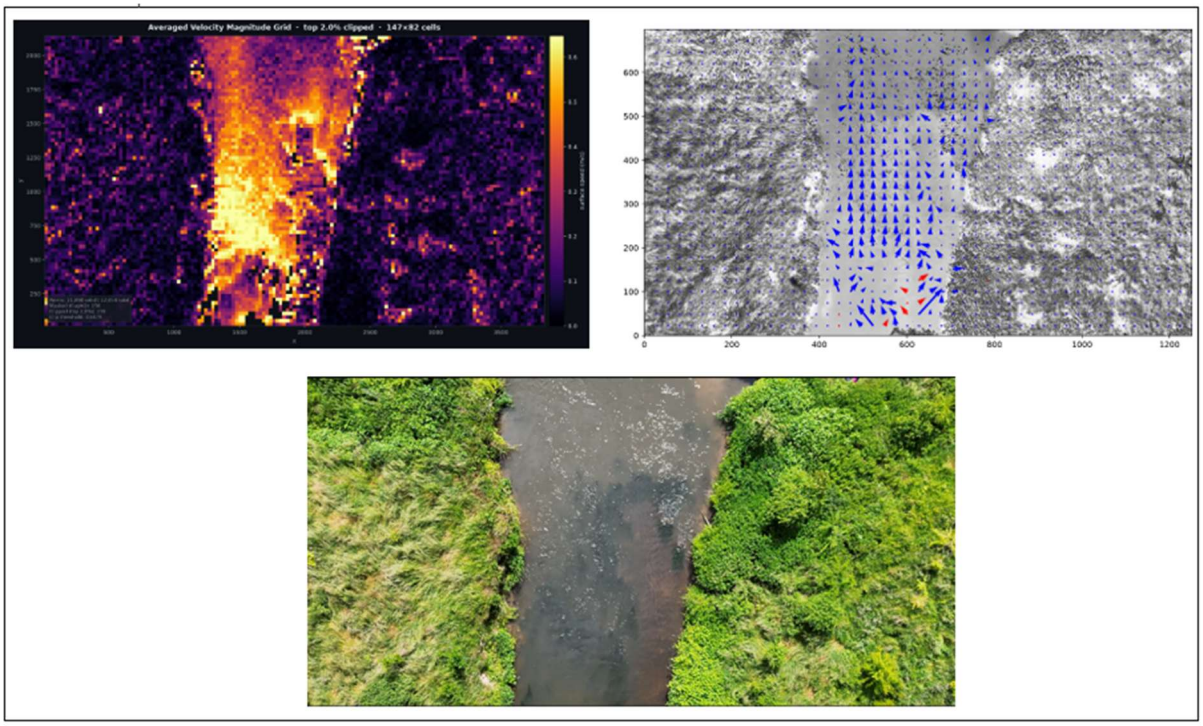


***Figure 9**. Location 1 at the Geul, selected based on the absence of obstacles and water and the easy access of the riverbank to enter with flowmeter. Water flows from south to north. **Top left:** Velocity grid shows speed in m/s where lighter colours in the centre of the river denote higher flow velocities. The dark purple reflects the riverbanks, in which brighter spots show moving trees. Image has a resolution of 4K with 40s length. **Top right:** Vector map, where the blue arrows show the direction of the flow. Increasing arrow size relates to higher flow velocities. Red arrows indicate locations where the algorithm detected flows deviating from the rest of the vector fields. Image has a resolution of 4K with 40s length. **Below:** RGB image.*

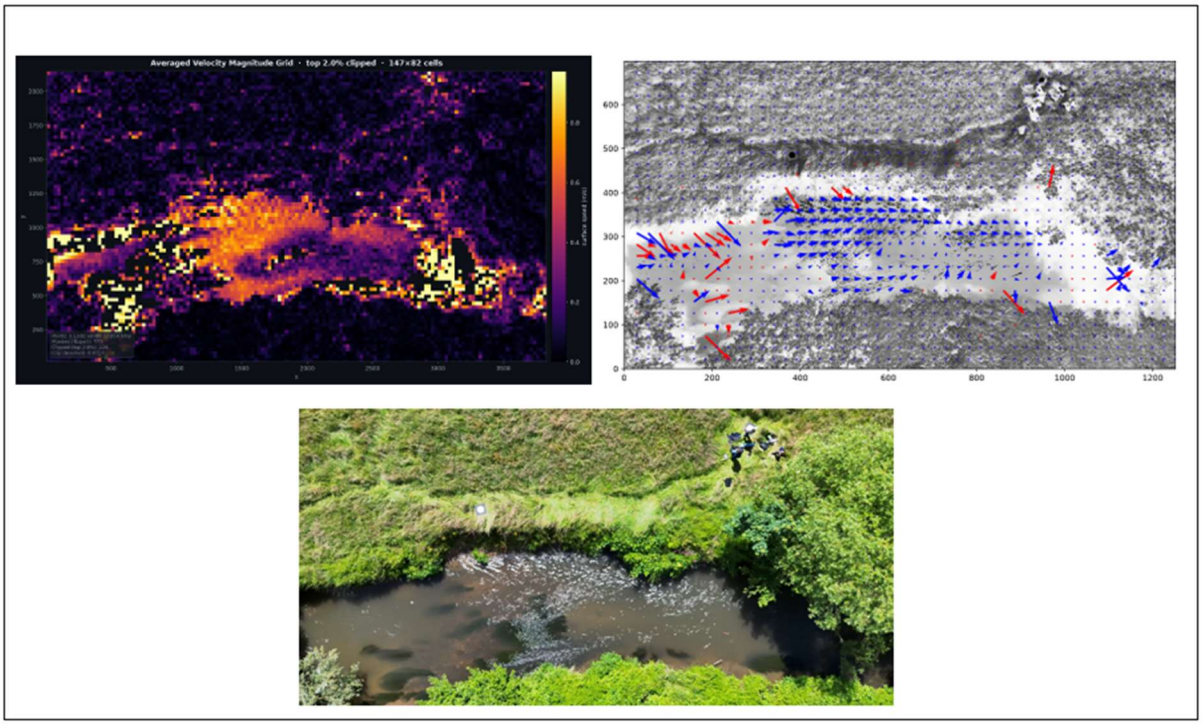


***Figure 10**. Location 2 at the Geul, selected based on the variance in flow velocities and minimal obstruction of shadow from trees. Water flows from west to east. **Top left:** Velocity grid shows speed in m/s where lighter colours at the edges of the river denote higher flow velocities. Image has a resolution of 4K with 40s length. **Top right:** Vector map, where the blue arrows show the direction of the flow. Increasing arrow size relates to higher flow velocities. Red arrows indicate locations where the algorithm detected flows deviating from the rest of the vector fields. Image has a resolution of 720p with 40s length. **Below:** RGB image.*

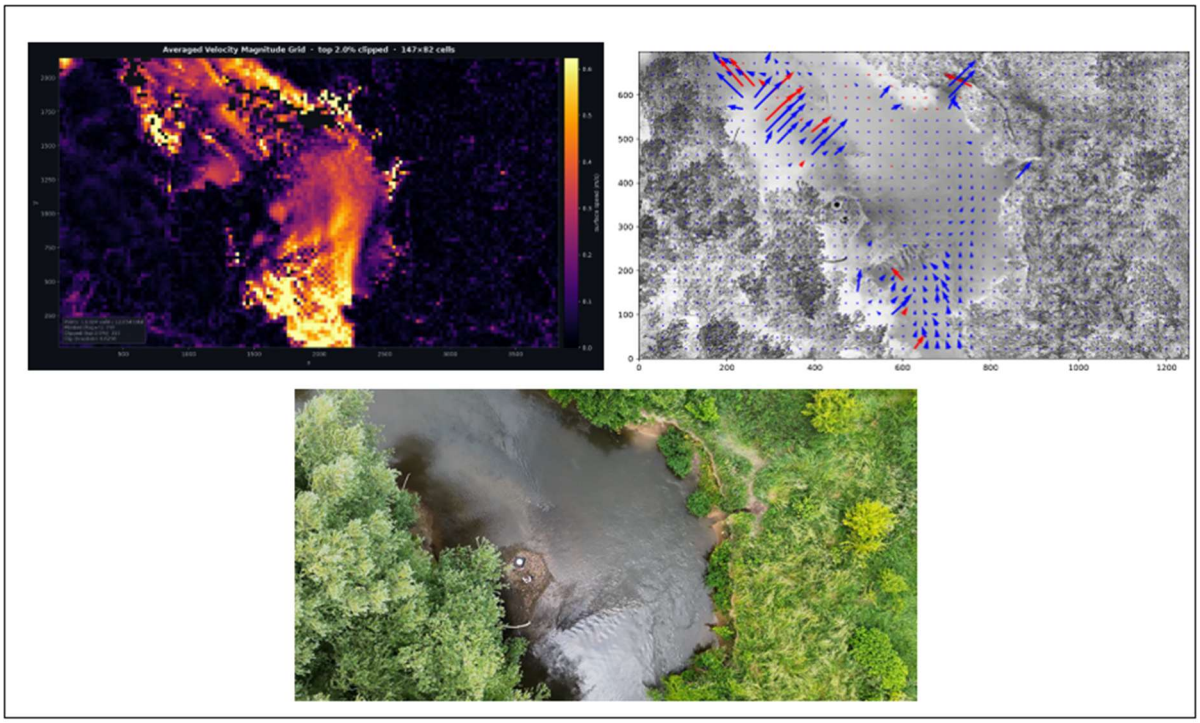


***Figure 11**. Location 3 at the Geul, selected for the presence of a bend with different flow dynamics compared to locations 1 and 2, and its easy access to the water for the validation data extraction. Water flows from southeast to northwest. **Top left:** Velocity grid shows speed in m/s where lighter colours upstream, near the inner bend of*

the river denote higher flow velocities, as well as the edges more upstream. Image has a resolution of 4K with 40s length. ***Top right:*** *Vector map, where the blue arrows show the direction of the flow. Increasing arrow size relates to higher flow velocities. Red arrows indicate locations where the algorithm detected flows deviating from the rest of the vector fields. Image has a resolution of 720 p with 40s length.* ***Below:*** *RGB image.*

The Waal has a less reliable method for validation data extraction, namely tracking physical object displacement from one side to the other side of the bridge. The values are plotted in Figure 12.

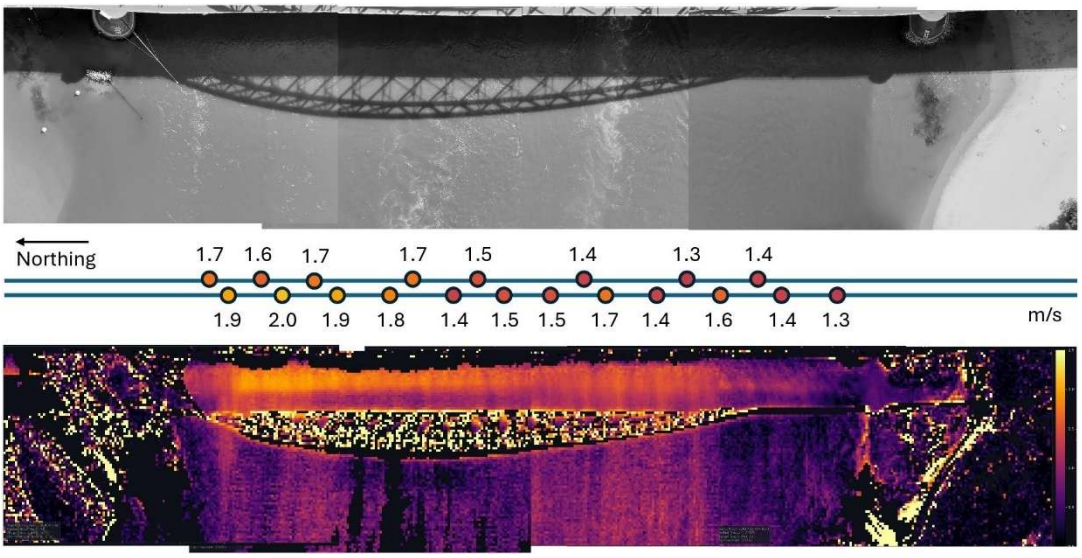


Figure 12. Validation points from the Waal distribution and its f validation points on the bridge plotted between the LSPIV velocity grid below and the RGB image above. Showing the velocity of the river compared to the validation points. Numbers at the validation points indicate the measured velocity, their colours correspond with the velocity grid below.

4.3 General Accuracy Assessment

The accuracy of assessment is performed on the 4k resolution, 40-second videos, as it was expected that this combination would have the highest chance to correctly estimate the velocity. A high resolution has a high pixel count and provides more spatial precision with which the displacement of particles can be tracked. Likewise, the altitude of 40 metres is relatively low, reducing the area covered per pixel. Meanwhile, a longer video includes more frame pairs for cross-correlation, decreasing noise from individual frames. As a result, these settings were expected to give the most reliable and flexible basis for the accuracy assessment. The impact of changing either flight altitude, video duration or video resolution will be discussed in following sections. The mean LSPIV velocity is calculated on the cells that contain the coordinates that were recorded using the RTK at the point of the flow velocity meter.

Table 1: Average results of the Geul and the Waal for the mean validation velocity, mean PIV velocity, Mean Absolute Error (MAE) and Relative Mean Absolute Error (RMAE). The results show

Location	Mean validation velocity (m/s)	Mean LSPIV velocity (m/s)	Mean Absolute Error (MAE):	Relative Mean Absolute Error (RMAE)
Geul Average	0.64	0.37	0.35	54.7%
Waal Average	1.54	0.97	0.58	37.39%

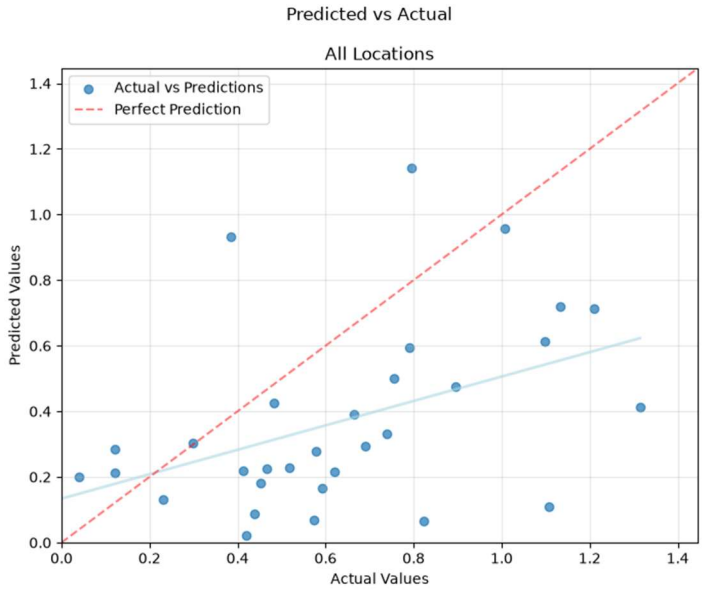


Figure 13. Predicted versus actual measurements of all validation points along the Geul river. The blue line indicates the trend of the predicted values, which is generally lower than the ideal prediction (red) line.

Mean error shows the bias in the estimation from the LSPIV method. It calculates the average difference between the drone measurements and the hand-measured validation data. It shows there is a systematic underestimation (negative bias). On average, the drone underestimates the river flow velocity by 0.27 m/s compared to validation data. This can be observed in Figure 13.

Shapiro-Wilk tests whether the dataset (specifically the distribution of the errors/differences) follows a normal distribution curve. With a p-value of 0.36, thus we do not reject the null hypothesis that the data is normally distributed. The errors are normally distributed, therefore parametric tests like the one-sample t-test can be used.

One-Sample T-Test is used to determine if the mean of the sample is significantly different from a known value, in this if there is zero error. The t-statistic is deeply negative (-4.5226) and the p-value is extremely small (8.9e-05). This proves that the -0.27 m/s underestimation is statistically significant.

Wilcoxon signed-rank test was used as a non-parametric equivalent to the one-sample t-test. It tests whether the median of the differences between paired observations skips away from zero, without assuming a normal distribution. This test backs up the t-test. The null hypothesis is the median difference between the LSPIV, and the validation measurements is zero. The resulting p-value of 1.1e-04 is significant and confirms that the median difference is significantly different from zero, reinforcing the evidence that underestimation is a systemic issue.

Lastly, the Pearson Correlation shows error as depend on flow velocity. It measures the strength and direction of a linear relationship between two continuous variables: the size of the LSPIV error and how fast the river was flowing. r = -0.63 shows a strong negative correlation. As the actual river flow velocity increases, the error becomes more negative (larger). The faster the river flows, the more the drone underestimates the flow velocity. The p-value is 1.4e-04 and is therefore a trend and not random noise.

4.4 Resolution Impact

In this section, the results of the impact of changing (downscaling) resolution are shown. The videos were down sampled versions of the original 45 second 4K video. Flight height is 40 metres. This analysis was performed on data from the Geul only, as the quality of the validation data is expected to be higher. In Figure 14, the resolutions are plotted against the MAE.

Table 2: Results of the influence of the resolution on mean LSPIV velocity (m/s), GSD (m/pixel) and mean absolute error (MAE). Compared against the mean validation velocity (m/s). The table shows a stable mean validation velocity among the resolutions, an increase in the mean LSPIV with a lower resolution (with the exception of 480p) and an increase in the mean absolute error when there is a lower resolution,

Resolution	Mean Validation Velocity (m/s)	Mean LSPIV Velocity (m/s)	GSD (m/pixel)	Mean Absolute Error (MAE):
3840 x 2160 (4k)	0.64	0.37	0.0095	0.35
1920 x 1080 (1080p)	0.66	0.57	0.0191	0.41
1280 x 720 (720p)	0.66	0.56	0.0286	0.42
854 x 480 (480p)	0.64	0.98	0.0429	0.80
640 x 360 (360p)	0.67	0.59	0.0572	0.64

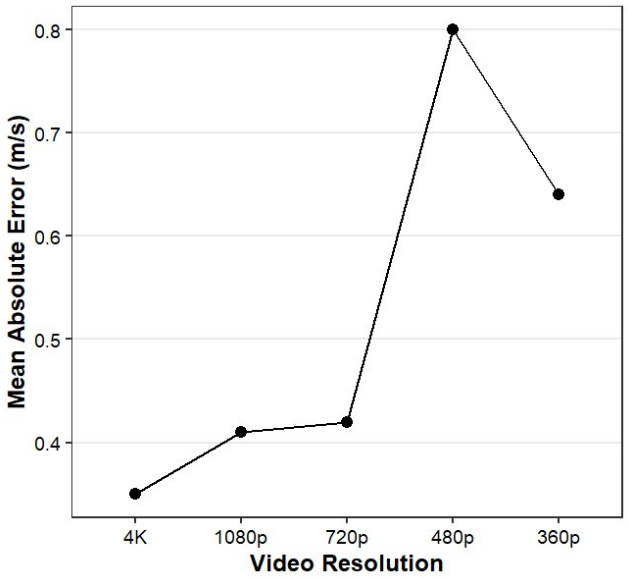


Figure 14. Shows the impact of input video resolution on the MAE. Note that the graph does not originate from 0 MAE. The MAE is lowest at 2160p resolution, at 0.35. 1080p and 720p have slightly worse results compared to 4k and similar between them. At 480p, there is a very large increase in MAE to 0.8. At 360p there is a slight decrease in MAE to 0.64.

4.5 Flight Altitude Impact

In this section, the results of the impact of changing flight altitude are shown. The video is recorded at 3840 by 2160 pixels resolution, however at different altitudes, this resulted in different GSDs at different altitudes. Due to time constraints, we were unable to fully validate and compare these results directly to validation data by matching their dynamic overlapping windows. Considering the range of validation flow velocity was between 1.32 and 1.98 m/s with a mean of 1.55 m/s, increasing altitude seems to close the gap between LSPIV measured Velocity and the validation data at the Waal.

Table 3. Results of the influence of the drone flight altitude above the Waal on the mean LSPIV velocity (m/s), GSD (m/pixel). The mean LSPIV velocity increases with altitude. The GSD changes accordingly with larger pixel size.

Altitude	Mean LSPIV Velocity (m/s)	GSD (m/pixel)
Waal, 40m	0.88	0.0095
Waal, 60m	0.96	0.0143
Waal, 120m	1.19	0.0286

4.6 Flight Duration Impact

In this section, the results of the impact of recording duration are shown. Resolution is 4K, flight height is 40 metres.

Table 4. Influence of duration of video-footage on the mean validation velocity (m/s), mean LSPIV velocity (m/s), GSD (m/pixels) and MAE on videos with a resolution of 4k, at flight height of 40 metres. Showing that the mean validation velocity and the mean LSPIV velocity are slightly variable. The mean absolute error increases when the duration of the footage decreases.

Duration	Mean Validation Velocity (m/s)	Mean LSPIV Velocity (m/s)	GSD (m/pixels)	Mean Absolute Error (MAE):
5 seconds	0.65	0.42	0.0095	0.37
10 seconds	0.65	0.41	0.0095	0.37
20 seconds	0.67	0.40	0.0095	0.36
30 seconds	0.65	0.41	0.0095	0.36
45 seconds	0.64	0.37	0.0095	0.35

4.7 Flight Observations

Wind speeds are based on observations from the KNMI, which are updated every 10 minutes. The nearest stations were selected as the reference point (KNMI, n.d.). These were *Maastricht-Aachen Aiport* for the Geul and *Volkel Airport* for the Waal. At the time of the flight at the Geul, there was an average windspeed of 5.1 m/s with a highest hourly average was 7.0 m/s around midday. The dominant wind direction was 230°. The cloud coverage was classified at 7 oktas (almost completely cloud covered). The drone flew at 40 metres above ground level.

There was an average windspeed of 2.7 m/s during the flights at the Waal, with a highest hourly average of 4.0 m/s around midday. The dominant wind direction was 33°. The cloud coverage was officially classified as unknown, but records show very sparse cloud coverage (1 okta). The drone flew

120m above ground level, except for flights numbers 7 and 8. There appears to be no difference in the horizontal and vertical drift magnitude for these flights.

The drone was set to hover at a specific location and altitude during data collection. Even under RTK position lock, the drone experienced some drifting, primarily due to wind. The resulting drift is shown below:

Table 5: average horizontal and vertical drift (cm) of the drone at the Geul and Waal. Showing that the Geul had more drift compared to the Waal.

Location / flight	Horizontal Drift (cm)	Vertical Drift (cm)
Geul Average	43,2	31,5
Waal Average	21,8	12,2
Average	26,9	17,5

5. Discussion

There is a systematic underestimation of the surface flow velocity compared to the validation data, revealed by the accuracy assessment. For the Geul, the mean velocity from the LSPIV-analysis (0.37 m/s) was considerably lower compared to the flow rate validation measurements (0.64 m/s), resulting in a MAE of 0.35 m/s (at 54.7% Relative Mean Absolute Error) and for the Waal MAE 0.58 (at 37.4% Relative Mean Absolute Error) (Table 1).

The statistical tests showed that the error is not caused by measurement noise but a systematic bias. The extent of the error scales with velocity, faster flows have a larger absolute error. Consequently, these results indicate that the pipeline's errors are not due to random variability but to a consistent deviation, related to the configuration parameters of the pipeline. Some parameters have been tested more extensively. The flight duration had a limited effect on accuracy (Table 4), while the resolution had an increasing mean absolute error once the video was downsampled (Table 2 and Figure 14). Table 3 shows LSPIV velocity estimations increase with flight altitude.
The following sections explore other literature studies, limitations that could (partly) explain the systematic bias and alternative approaches.

5.2 Comparison to other studies

Torres, W. et al. (2024) combined particle tracking with OpenPIV in the Antisana River (Ecuadorian Andes). Using lemon peels as particle trackers to accurately determine the speed of the water surface. Compared to the traditional Universal Current Meter F1 method, the LSPIV results showed a slight underestimation of the velocity. A similar study, also performed in Ecuador, from Ochoa-García et al. (2025) found similar results while researching the Yanuncay and Tarqui rivers. A strong correlation was found between LSPIV velocities when validated against the ground truth data from the HyQuest Redback current meter. However, the authors noted that wind can negatively affect the accuracy of flow evaluations. Lastly, a case study from Jyoti et al. (2023) on the Menomonee River in Wauwatosa found that drone-derived velocities are within 13-27% of their ground truth data collected with hand-held meters. However, the drone estimations are consistently lower than the validation data.

In our results, the flow velocity is also underestimated in most tested configurations. However, the errors are of much greater magnitude. The Pearson correlation between the error and the measured

flow velocity ($r = -0.63$, $p < 0.001$) indicates that this underestimation increases with flow velocity. A possible explanation is that at higher velocities, particles move further between frames, making it more likely they leave the interrogation window before the next frame is captured.

5.3 Limitations

Several limitations affect the interpretation of the results. Firstly, due to limited time, only two rivers could be captured using the drone. These rivers are not entirely representative of the problem being addressed: traversability of the military field by foot, vehicle or pontoon bridge. Both rivers were chosen based on the feasibility of the study, the time given and flight permissions. The Geul was chosen based on its velocity, as a relatively fast flowing river would be ideal to collect validation data from. The Waal was the fastest flowing river in the Netherlands with a width of at least 20 metres. This provides the study with only two rivers to represent the entire method, which is rather small, and the question is how representable these two rivers are for the purpose of the methodology.
Secondly, not all software options were applied. As the aim was to deploy, test and research one LSPIV algorithm and the choice had to be made early in the process, an extensive comparison of all available software was not possible. The open-source character of OpenPIV made it the basis of our python pipeline, while the robustness of RIVeR made it the preferred method for converting pixel size to physical units and for testing several videos in the GUI before processing them in the pipeline.

Furthermore, the current pipeline only compares consecutive frames, without the option of a frame gap. With a framerate of 30fps, the time between two consecutive frames is short, which results in small pixel displacements, especially in slow-flowing water. When this displacement gets close to the algorithm's sub-pixel accuracy limit, small measurement errors have a relatively larger effect, reducing the precision of the velocity estimate (Raffel et al., 1998). Using a frame gap would increase the displacement between compared frames and could improve the signal-to-noise ratio. A script for this was made (*check_parameters.py*), but due to time constraints this couldn't be used on tested on the data.

The validation data also brings some remarks. In the Geul, ground-truth data was obtained with a flow rate meter. The validation points were obtained across the river, at three pre-defined locations. For the Waal, the validation method was less accurate. Oranges were thrown from the Waalbrug, and the velocities were derived from the distance travelled under the bridge. A more robust approach could involve flow meters from a boat that can reach the middle of the river, or the use of an Acoustic Doppler Current Profiler (ADCP).

Next to being capable of capturing RGB footage, the drone also recorded thermal footage. On the original output, the raw thermal signal is encoded into a greyscale video drone, losing the actual thermal information. As a result, the thermal contrast of the water was non-detectible since it deviated too much from the riverbanks, making it unsuitable to derive flow velocities.

Research from Ochoa-García et al. (2025) already demonstrated that the wind speed can affect the outcomes. This factor was not reviewed separately in this study. Likewise with shadows, another factor that impacted the results. Near the Waalbrug, shadowed areas showed higher and more constant velocities than sun-illuminated areas in the resulting velocity maps. Along the sun-shadow boundary,

most resulting vectors are invalid. This suggests that the algorithm struggles to track movement in those conditions (Figure 15). The environmental conditions were part of the metadata but not (yet) researched more extensively (see section 3.2.3).
Moreover, the orthorectification and GSD calculation assume a fixed viewing geometry. With this, the camera-to-surface distance and therefore GSD remain constant across the image. It was only discovered during the late stage of our research that the drone footage of the Waal has been filmed at 86 degrees, rather than true nadir (90 degrees). As a result, the camera-to-surface distance is no longer constant across the image. Consequently, the GSD also varies slightly across the image, introducing additional uncertainty into the flow velocity estimates at this location. There was insufficient time to research a method to recalculate and correct this, meaning the results of the Waal must be interpreted with this small uncertainty in mind.

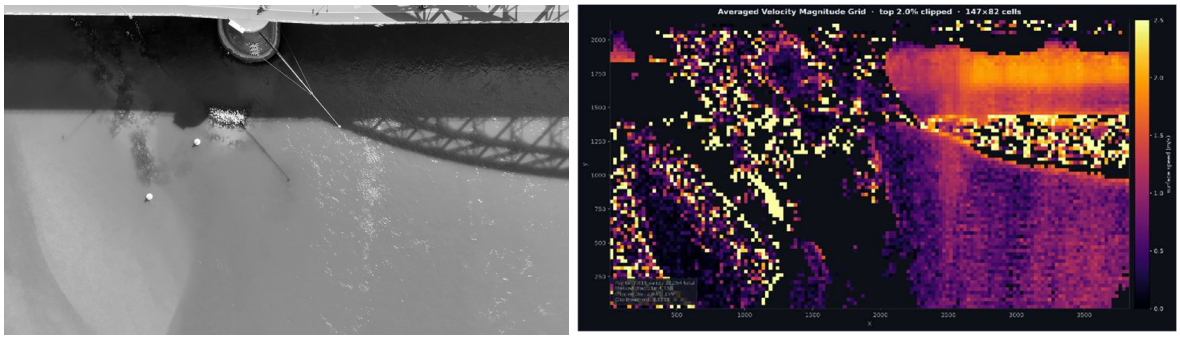


Figure 15. UAV footage of the Waal near the Waalbrug (left) and the corresponding velocity grid (right). The presence of a sandbank and the transition between shadowed and sunlit areas result in increased noise and unreliable velocity vectors in those regions.

In terms of the analysis, the availability of sufficient computational resources could be a constraint. Especially if the computing resources available in the military field are not compatible with processing longer high-resolution videos. The pipeline currently loads all data into memory during processing and RAM could reach its capacity, causing the pipeline to stall. Ideally, the workflow would clear processed data from the memory before each new process begins.

5.4 Potential alternatives

Next to OpenPIV, there are a lot of other models that could be used and tested for the same purpose. This study looked at the options and chose OpenPIV based on time and feasibility. This section will cover the most important alternatives.
Due to time restrictions, the application of deep learning models was deemed unfeasible. Deep learning models are, however, researched a lot nowadays and are an interesting approach for image-based flow measurement. Instead of correlating interrogation areas between 2 images, the approach of conventional PIV, deep-learning approaches can predict optical flow fields directly between consecutive frames. Literature studies often involve (a combination) of Convolutional Neural Networks (CNN). For example, Cai, S. (2019) described *LiteFlowNet*, a CNN that detects small, complex structures in the river. Chen, W. (2025) used a combination of CNN with long short-term memory and 3D CNN application built in Python. Mishra et al. (2024) demonstrated the potential of a CNN to predict surface flow velocities from video frames, using drone imagery of the Varuna River in India. Even though the CNN successfully predicted the river's maximum surface flow velocity, it overestimated velocity in areas with larger tracers. Suggesting that this approach has clear potential to enhance PIV analysis, but

deep learning models still require fine-tuning to distinguish between helpful and obstructive particles. Ansari, S. et al. (2023) developed a velocimetry scheme called *RivQNet*. The novel CNN algorithm robustly estimates surface velocity for different visibility, brightness, shade and resolutions.

A second alternative is the use of other wavelengths away from RGB. For instance, Bandini et al. (2020) explored the use of Surface Velocity Radar (SVR) mounted on a drone. The SVR detects reflected pulses for water surface velocity estimation, ranging from 0.8 to 15 m/s. Ground-truth data for validation was captured with an electromagnetic velocity probe (MFpro) and an Acoustic Doppler Current Profiler (ADCP). The SVR surveys were conducted on the largest river in Denmark and three rivers in Sweden. After processing with LSPIV-analysis, the resulting velocities matched well with the ground truth as velocity errors were 10 cm/s or less. While further research is needed for streams with varying roughness conditions, this method also shows considerable promise.

6. Conclusion

This study is a proof-of-concept for the combination of UAV footage in combination with an OpenPIV-based Python pipeline. The method showed to be feasible to produce velocity grids and vector fields from video footage, without the need for Ground Control Points. The method was tested on 2 rivers. For the Geul, the pipeline achieved 44.9% accuracy at 4K resolution at a length of 40 seconds. Statistical analysis revealed a systematic underestimation, quantified with a mean absolute error of 0.27 m/s, compared to the validation data. Mean velocities correspond well to validation data at lower resolutions, but the MAE increases at lower resolutions.

Results for the Waal should be considered low certainty, as the validation data collected at this location are of insufficient quality to support a direct and conclusive comparison with the LSPIV estimates. Their distribution and the acquisition method may not be representative for the Waal. However, results do show that the LSPIV velocity estimates are in the same order of magnitude as the validation data and share consistent mean values across the image.

Additionally, several operational conditions were tested in the field. First, several resolutions were tested by downscaling the imagery. The accuracy is highest for 4K resolution and diminishes with lower resolutions. At 360p the results become very unreliable, therefore a minimum resolution of 720p is recommended. This aspect is directly related to the flight altitude since the drone camera and thus pixels cover a wider area at higher altitude. The flight duration has limited influence on the accuracy as the MAE remains almost constant with different video lengths. Shorter hover times are recommended though, as they are more practical in conflict zones. A set minimum spatial resolution for detecting river flow velocity was not defined in this project because the definition of a ‘successful’ measurement is highly dependent on project goals. However, depending on the desired ground coverage or accuracy an indication of a minimum spatial resolution required can be estimated based on our results.

Several key constraints were identified, accompanied by recommendations to improve during future research. The drone recorded thermal imagery simultaneously with the RGB footage. Foremost produced no usable results, yet due to a lack of time no further analysis has been done to improve this imagery. As it has the advantage over RGB of being useful during nighttime conditions, it is

recommended to apply contrast-enhancing techniques to investigate the feasibility of the imagery. Secondly, the impact of wind on the surface flow velocity has not been quantified. Moreover, shadows from infrastructure or vegetation influence results negatively. Masking techniques can be applied to filter out outliers caused by these environmental factors. Additionally, this study described an alternative method to convert pixels to physical velocity units with the use of a multi-height drone experiment at a significant distance from the river. The GSD is a necessary component of the velocity determination, and for the military context the optimal method would not include any GCPs in the field. A method to extract it from the intrinsic camera parameters must be further investigated. Lastly, the presence of clear distinguishable particles in the water is necessary for the algorithm to be able to detect movement between two frames. Small errors might have been introduced due to small movements of the drone during hovering and not filming at true nadir over the Waal.
In short, the findings support the feasibility of UAV-based LSPIV as a remote alternative to in-situ flow velocity measurement, while emphasising that further refinement of the processing pipeline and data collection protocol is required before the method can be reliably deployed.

7. Recommendations

This study described the feasibility of estimating surface flow velocity using UAV videos as input for a Python-PIV pipeline. As a result, a proof-of-concept for the application of military traversability analysis is delivered. Building on the results of two rivers in the Netherlands, some recommendations have been listed below for further research and improvements.

7.1 Data acquisition

The thermal imagery recorded during fieldwork turned out to have insufficient contrast to be useful for PIV-analysis. The usage of thermal footage could be investigated further with contrast enhancement techniques, since velocity estimation from RGB footage would be unusable under nighttime conditions or conditions where a lot of shadow is visible. Likewise, the colour scales of the RGB imagery could be adjusted to create more contrast, so the algorithm would be even better able to detect movement. Additionally, automating the conversion of video codecs prior to processing would save significant manual labour and human errors. Currently, Handbrake is used to compress drone videos as the raw footage is incompatible with OpenPIV or RIVeR. Further development of the actual pipeline is also recommended. Right now, the pipeline requires significant computational resources, particularly memory usage for the processing of longer videos. Memory management would improve efficiency and practicality for fieldwork. In addition, not all configuration parameters of OpenPIV have been optimised yet, leaving room for exploration and improvement to the source code as well.

7.2 Data processing

Optimisation of parameters such as the search window size or implementing the possibility to vary the interval between compared frame pairs (a frame gap), might improve the results and reduce the observed error (Raffel et al., 1998). A script for optimising the interrogation window size and frame gap has already been made (check_parameters.py), but due to time constraints it was not possible to test this on the current data.
Additionally, the pipeline does not involve any deep learning methods. Studies have shown potential

for this approach (Mishra et al., 2024; Cai, S. et al., 2019; Ansari, S. et al., 2023; Chen, W. et al., 2025) but require more case studies and further development before they become reliable.

7.3 GSD

The GSD method as described in section 3.3.3 proved to be reliable, though it does require a calibration phase to derive the formula to convert pixel size to physical units. This requires GCPS to be placed in the field, which is undesirable in conflict zones, as personnel as to be deployed to place the physical markers. However, the GCPs do not have to be placed near the river, since the calibration can be carried out in a safe, controlled environment. Moreover, this can be done with any given drone-sensor combination if specific properties of the combination are known to derive the GSD-height relationship, as performed during this study. Consequently, this relationship is applied to imagery without having to place GCPs in the field at vulnerable locations. To further improve data acquisition and processing, future research should investigate methods to determine GSD without any physical markers, including the GCP-based calibration approach used in this study. This could be achieved by refining the use of integrated RTK positioning, intrinsic camera parameters (sensor size and focal length), water surface elevation, orientation data, and corrections for internal camera stabilization.

7.4 Validation

Currently, validation for the Geul and Waal was done using a flow rate meter and a physical object displacement, respectively. Future research should include a more robust and complete validation dataset that is representable for the rivers. The sample size should be larger with a more uniform distribution along the river width and in different flow conditions to assess under what conditions the methodology works. More accurate validation techniques, such as ADCP or boat-based flow meters, should be considered for better measurements of the local variability. Additionally, a complete cross-validation of resolution and video length combinations is recommended to see if shorter videos, which are preferred in conflict zones, do not deviate too much from the optimal trade-off between accuracy and computational accuracy.

7.5 Environmental factors

Environmental factors can influence the results during this study. These include the water depth, riverbank gradient, wind speed, and shadow. It is recommended to investigate their separate effects on the accuracy of velocity estimates in future research. For example, including a correction factor or performing several analyses with a variability in them to see if these cause the underestimation of the velocity.

7.6 Thermal data

The thermal data is lost during encoding when recording a thermal video using the DJI Matrice 4T. However, using thermal imagery could enable additional or alternative data in situations where RGB data may not be possible (such as data collection at night). To resolve this issue of encoding and loss of detail, we propose two focus points for further research: 1) Collect data using a sensor that can record the raw thermal data in video format. This would allow for more detailed processing because the data has not been encoded and is perfectly preserved. 2) Adjust the scaling of the image, either in the drone during recording if possible or in preprocessing after data collection. This could bring back

(some of) the detail that was lost by stretching the scaling in the lower range of temperatures which might bring back some contrast.

Acknowledgements

We would like to thank Blue Team Intelligence for this exciting challenge and interesting topic, with a special thanks to Jan van den berg for his ideas and critical questions. We would also like to thank our coach prof.dr.ir Lammert Kooistra for this guidance throughout this process. We are also grateful for Jelle ten Harkel, who as our expert helped us with technical questions. And lastly, we would like to thank Berry Onderstal for helping us collect the data with the drone.

Data Availability

All data and relevant output that was created will only be shared with Blue Team Intelligence and Wageningen University & Research. For more information regarding data-handling, please refer to Annex B, the Data Management Plan.
The script that was made and used can be found here:
https://github.com/eenoogzwaardvis/Drone-River-Flow-Velocity/tree/main

AI statement

During this project, the group has made use of the AI Claude (Opus 4.8 and Sonnet 4.6), Gemini (Flash 3.5) and DeepSeek (V3) to assist with:

- Writing and debugging code
- Explaining concepts
- Brainstorming
- Improving writing in documents

All AI-generated work was reviewed by its user.

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Appendix I

GSD calibration results of the multi-height experiment.

Flight height (m)	1080_pixel size (RIVeR) (m)	4K_encoded_pixelsize (RIVeR) (m)	Distance between GCPs (m)	GCP 1	GCP 2
30	0.02857	0.007092	8.24	4 (GCP midden)	3 (Hoekje gravel)
40	0.03734	0.009389	15.9	4 (GCP midden)	6 (GCP gras)
50	0.04688	0.01178	15.9	4 (GCP midden)	6 (GCP gras)
60	0.05651	0.01412	15.9	4 (GCP midden)	6 (GCP gras)
70	0.06675	0.01664	22.43	4 (GCP midden)	8 (GCP gras in buurt van brug)
80	0.07708	0.01918	22.43	4 (GCP midden)	8 (GCP gras in buurt van brug)
90	0.08738	0.02142	8.24	4 (GCP midden)	3 (Hoekje gravel)
100	0.09557	0.02388	15.9	4 (GCP midden)	6 (GCP gras)
110	0.1051	0.02636	15.9	4 (GCP midden)	6 (GCP gras)
120	0.1156	0.02861	22.43	4 (GCP midden)	8 (GCP gras in buurt van brug)

Appendix II

Configuration parameters, their description and function in the LSPIV analysis pipeline in Python

Configuration parameter	Description	Function in pipeline
FRAME_A_INDEX	First frame selected for PIV analysis. Takes index numbers.	The two consecutively selected frames are compared against each other to extract pixel displacement.
FRAME_B_INDEX	Second frame selected for PIV analysis. Takes index numbers.	
WINSIZE	Selects the window size of the interrogation window to subdivide frame A.	Determines the resolution in which frame A is processed.
SEARCHSIZE	Selects the search region of frame B in which the PIV looks for a match to the frame A interrogation window. Should be larger than WINSIZE.	Margin between the SEARCHSIZE and WINSIZE is the maximum particle displacement that can be detected.
OVERLAP	Determines how much overlap there is between two adjacent WINSIZE frames.	Determines spatial resolution of vector map and computation rate.
DT	Time (in seconds) between frame A and B. Must match the actual frame-to-frame timing of the camera used, otherwise the velocities will be scaled incorrectly.	Used to convert the pixel displacement to the surface flow velocities.
S2N_THRESHOLD	Threshold value for the determination whether a displacement vector is valid or invalid.	Determines the minimum acceptable signal-to-noise ratio (SNR) for a displacement vector to be considered valid.
REPLACE_METHOD	Method used to replace invalid vectors.	Determines how invalid vectors are replaced, affecting smoothness and noise level.
REPLACE_MAXITER	Number of times that the replace process is iterated.	Higher values result in more complete filling of invalid vectors, while lower values may leave gaps.
REPLACE_KERNEL	Neighbourhood used to replace invalid vectors.	Larger kernels create a smoother field, due to more averaging but may lead to over smoothing. Smaller kernels result in less smoothing, too small may fail to fill in invalid vectors.
GSD	Converts distances measured in pixels into distances measured in physical units. Must match calculated pixel size/GSD.	Used to convert the pixel displacement to the surface flow velocities.